Digit Independence under a Common Multiplier

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Abstract

For every integer base $p \geq 2$ and every finite set S of distinct positive natural numbers, we construct a single multiplier K such that the infinite, zero-padded base-p digit vectors of the numbers aK, with $a \in S$, are linearly independent over $\mathbb{Z}/p\mathbb{Z}$. The base need not be prime. The proof constructs a unit lower-triangular matrix from threshold-crossing digits and then realizes its rows as linear projections of digit vectors produced by one block-packed multiplier. A Lean formalization accompanies this note.

1 Digit vectors and the theorem

Fix a natural-number base $p \geq 2$. For $N, r \in \mathbb{N}$, define the rth digit of N, viewed in $\mathbb{Z}/p\mathbb{Z}$, by

$$d_{p,r}(N) = \left[\left\lfloor \frac{N}{p^r} \right\rfloor \right]_p \in \mathbb{Z}/p\mathbb{Z}. \tag{1}$$

Equivalently, this is the ordinary rth base-p digit $\lfloor N/p^r \rfloor \bmod p$, followed by the canonical map into $\mathbb{Z}/p\mathbb{Z}$.

The infinite, zero-padded digit vector is

$$D_p(N) = (d_{p,0}(N), d_{p,1}(N), d_{p,2}(N), \ldots) \in (\mathbb{Z}/p\mathbb{Z})^{\mathbb{N}}. \tag{2}$$

Theorem 1 (Digit independence under a common multiplier)**.** *Let $p \geq 2$, and let $S \subset \mathbb{N}_{>0}$ be finite. There exists $K \in \mathbb{N}$ such that the family*

$$\big(D_p(aK)\big)_{a \in S}$$

is linearly independent over $\mathbb{Z}/p\mathbb{Z}$.

Pairwise distinctness is built into the fact that S is a set. No primality assumption is made. Thus Theorem 1 includes composite p, even though $\mathbb{Z}/p\mathbb{Z}$ then has zero divisors.

In Lean the family is indexed by the subtype of elements of a `Finset Nat`. The condition that zero does not belong to the finset expresses positivity. The main theorem is named `DigitIndependence.exists_digits_linearIndependent`.

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2 Proof

Write

$$A = \sum_{a \in S} a, \qquad m = A^2 + 1, \qquad P = p^m. \tag{3}$$

Because $p \geq 2$, the elementary estimate $x < 2^{x+1}$ gives

$$A^2 < P. \tag{4}$$

If S is empty, the assertion is immediate. The construction below also handles that case formally. We therefore proceed with the estimates needed for elements of S.

2.1 First threshold crossings

For every $i \in S$, let q_i be the least natural number for which

$$P \leq iq_i. \tag{5}$$

Such a number exists because $i > 0$. Minimality yields

$$i(q_i - 1) < P \leq iq_i. \tag{6}$$

Moreover $q_i \leq P$, because $i \geq 1$ implies $P \leq iP$.

2.2 A unit lower-triangular matrix

Define a square matrix indexed by S:

$$B_{a,i} = d_{p,m}(aq_i) - d_{p,m}(a(q_i - 1)) \in \mathbb{Z}/p\mathbb{Z}. \tag{7}$$

We first compute its diagonal. From Equation (6),

$$P \leq iq_i = i(q_i - 1) + i < P + i < 2P.$$

The final inequality follows from $i \leq A < P$. Since $P = p^m$, division by p^m shows that

$$d_{p,m}(iq_i) = 1, \qquad d_{p,m}(i(q_i - 1)) = 0.$$

Consequently,

$$B_{i,i} = 1. \tag{8}$$

Now suppose $a < i$. From Equation (6) we have $iq_i < P + i$. Using commutativity of multiplication,

$$\begin{aligned} i(aq_i) &= a(iq_i) \\ &< a(P + i) \\ &< (a + 1)P \\ &\leq iP. \end{aligned}$$

The strict middle inequality is equivalent to $ai < P$, which follows from $a, i \leq A$ and Equation (4). The last inequality follows from $a + 1 \leq i$. Cancelling the positive factor i gives

$$aq_i < P = p^m. \tag{9}$$

The same bound holds for $a(q_i - 1)$. Hence both digits in Equation (7) vanish, and

$$B_{a,i} = 0 \qquad \text{whenever } a < i. \tag{10}$$

After ordering S by the usual order on $\mathbb{N}$, Equations (8) and (10) say that B is lower triangular with every diagonal entry equal to one. Therefore

$$\det B = 1. \tag{11}$$

This equality is essential for composite bases. Over a ring with zero divisors, a merely nonzero determinant need not imply invertibility. Here the determinant is exactly 1, hence a unit. It follows that B is invertible and its rows are linearly independent over $\mathbb{Z}/p\mathbb{Z}$.

2.3 Packing all threshold multipliers into one number

It remains to replace the separate multipliers q_i by one common multiplier K. Choose an enumeration

$$S = \{s_0, \dots, s_{n-1}\}.$$

Choose a block length L such that

$$m < L, \qquad aq_i < p^L \quad \text{for all } a, i \in S. \tag{12}$$

The Lean proof uses the explicit, deliberately generous value

$$L = AP + m + 1. \tag{13}$$

Indeed $a \le A$ and $q_i \le P$, so $aq_i \le AP < p^L$.

Pack both q_i and $q_i - 1$ into disjoint blocks of L base-p digits:

$$K = \sum_{j=0}^{n-1} q_{s_j} p^{jL} + \sum_{j=0}^{n-1} (q_{s_j} - 1) p^{(n+j)L}. \tag{14}$$

Multiplication distributes over this expression:

$$aK = \sum_{j=0}^{n-1} (aq_{s_j}) p^{jL} + \sum_{j=0}^{n-1} a(q_{s_j} - 1) p^{(n+j)L}.$$

Condition (12) ensures that every coefficient fits in its own block. Therefore no carry crosses a block boundary, and

$$d_{p,jL+m}(aK) = d_{p,m}(aq_{s_j}), \tag{15}$$

$$d_{p,(n+j)L+m}(aK) = d_{p,m}(a(q_{s_j} - 1)). \tag{16}$$

Define a linear coordinate projection

$$T : (\mathbb{Z}/p\mathbb{Z})^{\mathbb{N}} \longrightarrow (\mathbb{Z}/p\mathbb{Z})^S \tag{17}$$

by

$$(Tx)_{s_j} = x_{jL+m} - x_{(n+j)L+m}. \tag{18}$$

Equations (7), (15), and (16) give

$$T(D_p(aK)) = B_{a,\bullet}, \tag{19}$$

where $B_{a,\bullet}$ denotes the row of B indexed by a.

The rows of B are linearly independent. If the original vectors $D_p(aK)$ satisfied a linear relation, applying T would give the same relation among the rows of B. All its coefficients must therefore vanish. This proves Theorem 1. □

Remark 1 (A finite certificate)**.** *Although the digit vectors are modeled as functions* $\mathbb{N} \to \mathbb{Z}/p\mathbb{Z}$*, the proof reads only* $2|S|$ *carefully selected coordinates. Thus the construction provides a finite certificate of linear independence.*

3 Relation to Mathlib's standard digit representation

The formalization uses the convenient $\mathbb{Z}/p\mathbb{Z}$-valued definition

$$d_{p,r}(N) = [\lfloor N/p^r \rfloor]_p .$$

Mathlib's standard `Nat.digits` function returns a finite list of natural-number digits. After extending that list by zeros and casting its rth entry to $\mathbb{Z}/p\mathbb{Z}$, the two definitions agree, because canonical casting to $\mathbb{Z}/p\mathbb{Z}$ takes the residue modulo p. The relevant library statement is `Nat.getD_digits`. A bridge lemma between these presentations is a natural addition to the public API, allowing later results to be rewritten in either form without changing the proof construction.

4 Structure of the Lean formalization

The formalization is contained in `DigitIndependence(1).lean`. Its main components are:

- `digit` and `digits`: one digit and the infinite zero-padded digit vector;
- `pack` and `packFin`: block packing for lists and finite tuples;
- `digit_add_high_block` and `digit_tail_block`: isolation and shifting of base-p blocks;
- `digit_pack_get` and `digit_packFin`: extraction of a digit from a selected block;
- `mul_pack` and `mul_packFin`: compatibility of packing with multiplication;
- `exists_crossing`: existence of the least threshold multiplier q_i;
- `exists_digits_linearIndependent`: Theorem 1 for every $p \geq 2$;
- `exists_digits_linearIndependent_of_prime`: compatibility corollary for prime bases.

In the last step, the Lean proof constructs the linear map T from Equation (18). The rows of the triangular matrix are the images of the full digit vectors under T. The theorem `LinearIndependent.of_comp` then lifts linear independence from the images to the original family. To cover composite p, the proof applies `Matrix.linearIndependent_cols_of_isUnit` to the transpose of B.

5 Checking the file

Place the Lean file in a project depending on a compatible recent version of mathlib and run

```
lake env lean 'DigitIndependence(1).lean'.
```

Before archival or publication, the exact Lean and mathlib versions should be pinned in `lean-toolchain` and `lakefile.toml`.

6 Current limitations and possible extensions

- The bounds for m, L, and therefore K were chosen for a short elementary proof, not for efficiency.
- The theorem covers every base $p \geq 2$, but still uses $\mathbb{Z}/p\mathbb{Z}$ as the coefficient ring. Separating the digit base from the coefficient ring would require a different public formulation.
- The construction is explicit, but the current theorem states existence rather than exposing an efficient executable algorithm.
- Natural extensions include sharper bounds, a standard `Nat.digits` corollary, independent base and coefficient ring, and a formalized finite-interpolation consequence.

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